



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Friday, 21 August 2026, 4:02 PM**State** Finished**Completed on** Friday, 21 August 2026, 4:06 PM**Time taken** 4 mins 19 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main()
3  {
4      int num;
5      scanf("%d",&num);
6      int i;
7      int count=0;
8      for(i=1;i<=num;i++)
9      {
10         count++;
11         if(num%i==0)
12         {
13             //printf("%d ",i);
14             count++;
15         }
16         count++;
17     }
18     count++;
19     printf("%d",count);
20 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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